

Sri Lanka Institute of Information Technology



Streetify

Web-Based Transportation System

Assignment Part 01 Enhanced Entity Relationship Diagram Part B - EER Diagram Design

Group ID : 2026-Y2-S1-KU-24 (Group 24)

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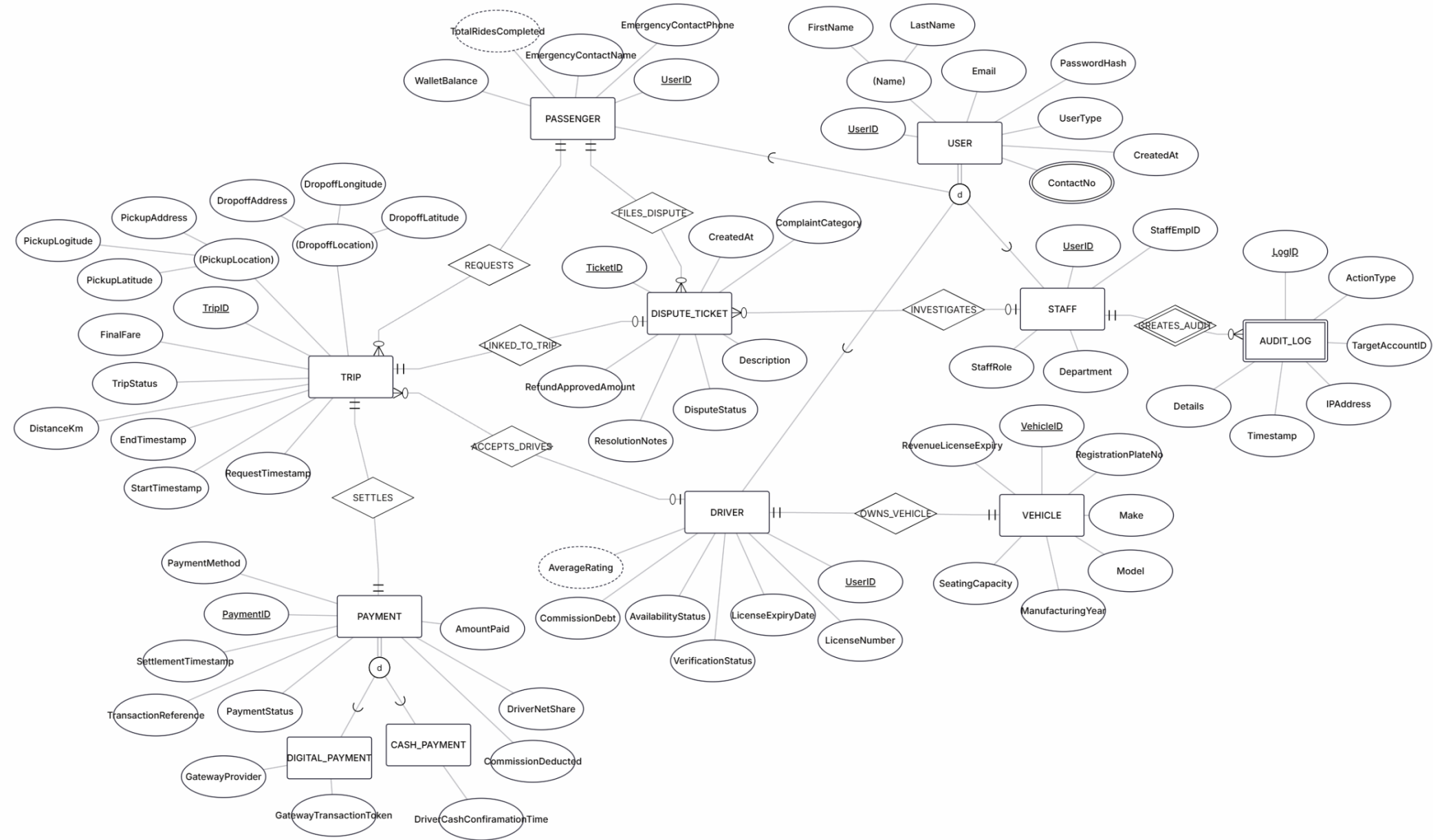
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Streetify EER Diagram Component & Structural Mapping

Category	Component	Type	Cardinality	Details & Attributes
Supertype	USER	Strong Entity	-	UserID (PK), Name (Composite: FirstName, LastName), Email, PasswordHash, UserType, CreatedAt, ContactNo (Multivalued)
Subtypes (ISA)	PASSENGER	Specialization	Inherits PK	EmergencyContactName, EmergencyContactPhone, WalletBalance, /TotalRidesCompleted (Derived)
	DRIVER	Specialization	Inherits PK	LicenseNumber, LicenseExpiryDate, VerificationStatus, AvailabilityStatus, CommissionDebt, /AverageRating (Derived)
	STAFF	Specialization	Inherits PK	StaffEmpID, Department, StaffRole
Hierarchy Constraints	USER → Subtypes	ISA Hierarchy	Disjoint (<i>d</i>), Total	Double line; every user must be exactly one subtype role (PASSENGER, DRIVER, or STAFF).
Core Entities	VEHICLE	Strong Entity	-	VehicleID (PK), RegistrationPlateNo, Make, Model, ManufacturingYear, SeatingCapacity, RevenueLicenseExpiry
	TRIP	Strong Entity	-	TripID (PK), PickupLocation (Composite), DropoffLocation (Composite), RequestTimestamp, StartTimestamp, EndTimestamp, DistanceKm, TripStatus, FinalFare

	PAYMENT	Strong Entity	-	PaymentID (PK), PaymentMethod, AmountPaid, CommissionDeducted, DriverNetShare, PaymentStatus, TransactionReference, SettlementTimestamp
	DISPUTE_TICKET	Strong Entity	-	TicketID (PK), ComplaintCategory, Description, DisputeStatus, ResolutionNotes, RefundApprovedAmount, CreatedAt
Weak Entity	AUDIT_LOG	Weak Entity	-	LogID (<i>Partial Discriminator</i>), ActionType, TargetAccountID, IPAddress, Timestamp, Details
Core Relationships	OWNS_VEHICLE	Binary	1 : 1 (Total : Total)	DRIVER (1,1) ↔ VEHICLE (1,1)
	REQUESTS	Binary	1 : N (Partial : Total)	PASSENGER (0,N) ↔ TRIP (1,1)
	ACCEPTS_DRIVES	Binary	1 : N (Partial : Partial)	DRIVER (0,N) ↔ TRIP (0,1)
	SETTLES	Binary	1 : 1 (Total : Total)	TRIP (1,1) ↔ PAYMENT (1,1)
	FILES_DISPUTE	Binary	1 : N (Partial : Total)	PASSENGER (0,N) ↔ DISPUTE_TICKET (1,1)
	LINKED_TO_TRIP	Binary	1 : N (Partial : Total)	TRIP (0,1) ↔ DISPUTE_TICKET (1,1)
	INVESTIGATES	Binary	1 : N (Partial : Partial)	STAFF (0,N) ↔ DISPUTE_TICKET (0,1)
Identifying Relationship	CREATES_AUDIT	Identifying (Double Diamond)	1 : N (Partial : Total)	STAFF (0,N) ↔ AUDIT_LOG (1,1)
Recursive Relationship	SUPERVISES	Unary / Recursive	1 : N (Partial : Partial)	STAFF (Supervisor) (0,1) ↔ STAFF (Subordinate) (0,N)
Aggregation	TRIP_SETTLEMENT	Aggregation	1 : 1	Aggregates [TRIP + PAYMENT] as a single object linked to DISPUTE_TICKET.

Enhanced Entity Relationship Diagram



Proofs of Enhanced Entity Relationship Diagram Drawing

The screenshot displays the ERDPlus web application interface. The main workspace shows a complex ERD for a system named "EER_Group24". The diagram includes entities such as DRIVER, VEHICLE, and OWNS_VEHICLE, with various attributes and relationships. A sidebar on the right allows editing the "OWNS_VEHICLE" relationship, showing options for identifying, exact constraints, and cardinalities. The bottom of the interface shows a toolbar with options like Entity, Label, Delete, Undo, Redo, Connect, Save, Restore, and Download image.

Diagram name: EER_Group24

Relationship: OWNS_VEHICLE

Identifying: ☐

Exact constraints: ☐

Edit cardinalities for: DRIVER

Min Cardinality:

Max Cardinality:

Add attribute

Advertisement

ERDPlus Diagrams SQL

Learn About

Diagram name: EER_Group24

Entity Label Delete Undo Redo Connect Save Restore Download image

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